

MD REHAN

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Education

- **B.Tech in Computer Science & Engineering** — VIT Bhopal University **2024 – 2028**
- **Class XII (ISC)** — St. Joseph & Mary's School, Kolkata **2024**
- **Class X (ICSE)** — St. Joseph & Mary's School, Kolkata **2022**

Skills

- **Languages & Frontend:** Python, C++, JavaScript, HTML5, CSS3, React.js, Tailwind CSS
- **Backend & Databases:** Node.js, Express.js, FastAPI, REST APIs, SQL, MongoDB, Prisma
- **AI/ML & Data:** LangChain, LangGraph, RAG, Scikit-learn, NumPy, Pandas, Matplotlib
- **DevOps & System Concepts:** Docker, Apache Kafka, Git & GitHub, System Design, JWT Auth, Microservices, API Design

Academic Projects

SmartFarm Console & AgroBot  | *Next.js 16, Tailwind CSS v4, PostgreSQL, Prisma, Docker, LangGraph, Gemini 2.5 Flash*

- Engineered an accessibility-first agricultural platform for crop diagnostics, leaf pathology scanning, and real-time weather notifications.
- Developed a LangGraph-powered AgroBot assistant featuring dynamic tool calling, multilingual voice chat in 6 regional languages, and offline ML models.
- Built a robust responsive interface using Next.js 16, PostgreSQL (via Prisma ORM), and Docker containerization to support remote farming workloads.

LectureMind AI  | *React 19, Vite, Tailwind CSS, Web Speech API, Node.js, Express, FastAPI, PostgreSQL, Prisma, LangGraph, RAG*

- Developed an intelligent study workspace that automatically processes lecture recordings, PDF notes, and videos into structured study guides.
- Architected stateful LangGraph workflows to ingest documents, generate hierarchical concept maps, and implement Socratic voice diagnostics.
- Implemented a hybrid Token/Vector RAG using sliding-window chunking with Gemini Embeddings and the SuperMemo-2 (SM-2) algorithm for active recall.

Welth AI  | *Next.js 15, React 19, Tailwind CSS v4, Supabase, Prisma ORM, Inngest, Gemini AI, LangChain, Arcjet, Clerk*

- Created a premium personal finance tracker and AI wealth manager featuring OCR receipt logging and localized budget recommendations.
- Integrated AI Vision OCR receipt scanning with Gemini 2.5 Flash, extracting vendors, tax parameters, and items with sub-second latency.
- Built a RAG-driven financial coach for automated budget auditing and integrated background scheduler jobs using Inngest event queues and Resend.

Prept – AI Interview Coach  | *Next.js 16, React 19, Tailwind CSS v4, Shadcn UI, Python FastAPI, PostgreSQL, Supabase, Prisma, Arcjet*

- Engineered an AI-powered mock interview prep platform with interactive simulated coding and behavioral feedback.
- Designed an ATS Resume Analyzer to scan uploads against target job descriptions, identifying skill gaps and suggesting bullet point optimizations.
- Formulated custom lightweight JWT edge session authentication running within Next.js Middleware, secured with Arcjet rate-limiting and WAF.

Certifications

- **The Bits and Bytes of Computer Networking** – Coursera — Google Professional Certificate
- **Marketing Analytics** – NPTEL — National Programme on Technology Enhanced Learning
- **Cloud Computing** – NPTEL — National Programme on Technology Enhanced Learning